80386DX Architecture & Functional Units MPCA Module 5

> **Definition:** The **Intel 80386DX** is a true **32-bit microprocessor** with a 32-bit internal architecture, a 32-bit Data Bus, and a 32-bit Address Bus, capable of addressing **4 GB of physical memory** and up to **64 Terabytes (TB) of virtual memory**.

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1. Detailed Technical Explanation

1. Functional Block Diagram of 80386DX

The 80386DX architecture is organized into **three major sections** containing six distinct functional units:

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80386

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## 2. Address Spaces in 80386DX

1. **Physical Address Space: 2^32 = 4 GB** of physical RAM directly addressable.

2. **Vi

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## 3. Core Concepts & Memory Keywords

- **32-bit Data & Address Bus:** Transfers 32-bit dwords in a single 2-clock bus cycle.

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4. Must-Write Points for Exams

- The 80386DX has a 32-bit data bus and 32-bit address bus (unlike 80386SX which has a 16-bit data bus and 24-bit address bus).

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5. Quick Recall Flow

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80386
64TB
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